IVOR Paine Memorial Hospital

Database Table Descriptions

24I-0809 | 24I-0531 | 24I-0682

Ward

Represents a hospital ward, the top-level physical division of the hospital.

Columns

Column Name	Data Type	Nullable	Description
wardName	VARCHAR(50)	NOT NULL	Unique name identifying the ward

Primary Key

PK: wardName

Foreign Keys

None

Constraints

- PK_Ward — PRIMARY KEY (wardName)

CareUnit

Represents a care unit within a ward. Patients are grouped into care units, each overseen by a staff nurse.

Columns

Column Name	Data Type	Nullable	Description
unitNo	INT	NOT NULL	Hospital-wide unique care unit number
ward	VARCHAR(50)	NOT NULL	Ward this care unit belongs to

Primary Key

PK: unitNo

Foreign Keys

- ward → Ward(wardName)

Constraints

- PK_CareUnit — PRIMARY KEY (unitNo)
- FK_CareUnit_Ward — FOREIGN KEY (ward) REFERENCES Ward(wardName) ON UPDATE CASCADE ON DELETE CASCADE

Bed

Represents a hospital bed with a hospital-wide unique number, belonging to a specific ward. Each bed can be occupied by at most one patient.

Columns

Column Name	Data Type	Nullable	Description
bedNo	INT	NOT NULL	Hospital-wide unique bed number
ward	VARCHAR(50)	NOT NULL	Ward this bed is located in

Primary Key

PK: bedNo

Foreign Keys

- ward → Ward(wardName)

Constraints

- PK_Bed — PRIMARY KEY (bedNo)
- FK_Bed_Ward — FOREIGN KEY (ward) REFERENCES Ward(wardName) ON UPDATE CASCADE ON DELETE CASCADE

Staff

Supertype entity for all medical staff. Both Consultant and Doctor are subtypes that inherit staffID and name from this table.

Columns

Column Name	Data Type	Nullable	Description
staffID	VARCHAR(10)	NOT NULL	Unique staff identifier (e.g. C001, D001)
name	VARCHAR(100)	NOT NULL	Full name of the staff member

Primary Key

PK: staffID

Foreign Keys

None

Constraints

- PK_Staff — PRIMARY KEY (staffID)

Consultant

Represents a consultant — a senior doctor with a defined specialty who leads a team of doctors. Subtype of Staff (EER disjoint specialisation).

Columns

Column Name	Data Type	Nullable	Description
staffID	VARCHAR(10)	NOT NULL	PK and FK to Staff — shared identity
specialty	VARCHAR(100)	NOT NULL	Medical specialty (e.g. Cardiology, Orthopedics)

Primary Key

PK: staffID

Foreign Keys

- staffID → Staff(staffID)

Constraints

- PK_Consultant — PRIMARY KEY (staffID)
- FK_Consultant_Staff — FOREIGN KEY (staffID) REFERENCES Staff(staffID) ON DELETE CASCADE ON UPDATE CASCADE

Doctor

Represents a doctor who belongs to a consultant's team and is responsible for patient care. Subtype of Staff. Position indicates seniority level.

Columns

Column Name	Data Type	Nullable	Description
staffID	VARCHAR(10)	NOT NULL	PK and FK to Staff — shared identity
position	VARCHAR(20)	NOT NULL	Seniority level: student | jh | sh | ar | r
consultant	VARCHAR(10)	NULL	Current consultant this doctor reports to

Primary Key

PK: staffID

Foreign Keys

- staffID → Staff(staffID)
- consultant → Consultant(staffID)

Constraints

- PK_Doctor — PRIMARY KEY (staffID)
- FK_Doctor_Staff — FOREIGN KEY (staffID) REFERENCES Staff(staffID)
- CK_Doctor_Position — CHECK (position IN ('student','jh','sh','ar','r'))

DoctorTeam

Records the assignment of a doctor to a consultant's team, including the date the doctor joined. Implements the 'leads a team of' relationship from the ERD.

Columns

Column Name	Data Type	Nullable	Description
doctor	VARCHAR(10)	NOT NULL	FK to Doctor — the team member
consultant	VARCHAR(10)	NOT NULL	FK to Consultant — the team leader
dateJoined	DATE	NOT NULL	Date the doctor joined this consultant's team

Primary Key

PK: (doctor, consultant)

Foreign Keys

- doctor → Doctor(staffID)
- consultant → Consultant(staffID)

Constraints

- PK_DoctorTeam — PRIMARY KEY (doctor, consultant)
- FK_DoctorTeam_Doctor — FOREIGN KEY (doctor) REFERENCES Doctor(staffID)
- FK_DoctorTeam_Consultant — FOREIGN KEY (consultant) REFERENCES Consultant(staffID)

PastExperience

Stores a doctor's prior employment history. Weak entity — each record is meaningless without its owning Doctor. A doctor may have multiple experience entries.

Columns

Column Name	Data Type	Nullable	Description
doctorID	VARCHAR(10)	NOT NULL	Partial PK + FK to Doctor
expID	INT	NOT NULL	Partial PK — sequential per doctor
dateStart	DATE	NOT NULL	Start date of this employment
dateEnd	DATE	NULL	End date (NULL if still active)
position	VARCHAR(50)	NOT NULL	Role held at the establishment

Column Name	Data Type	Nullable	Description
establishment	VARCHAR(150)	NOT NULL	Name of the hospital or institution

Primary Key

PK: (doctorID, expID)

Foreign Keys

- doctorID → Doctor(staffID)

Constraints

- PK_PastExperience — PRIMARY KEY (doctorID, expID)
- FK_PastExp_Doctor — FOREIGN KEY (doctorID) REFERENCES Doctor(staffID) ON DELETE CASCADE

PerformanceGrade

Records performance grades assigned to a doctor by their consultant every six months. Weak entity — grades have no meaning without their owning Doctor.

Columns

Column Name	Data Type	Nullable	Description
doctorID	VARCHAR(10)	NOT NULL	Partial PK + FK to Doctor
perfID	INT	NOT NULL	Partial PK — sequential per doctor
grade	VARCHAR(5)	NOT NULL	Performance grade (e.g. A, B+, C)
dat	DATE	NOT NULL	Date the grade was assigned

Primary Key

PK: (doctorID, perfID)

Foreign Keys

- doctorID → Doctor(staffID)

Constraints

- PK_PerformanceGrade — PRIMARY KEY (doctorID, perfID)
- FK_PerfGrade_Doctor — FOREIGN KEY (doctorID) REFERENCES Doctor(staffID) ON DELETE CASCADE

Nurse

Supertype entity for all nurses. Subtypes are Sister, StaffNurse, and NonRegisteredNurse. Every nurse works on exactly one ward and has a defined role.

Columns

Column Name	Data Type	Nullable	Description
nurseID	VARCHAR(10)	NOT NULL	Unique nurse identifier
name	VARCHAR(100)	NOT NULL	Full name of the nurse
role	VARCHAR(20)	NOT NULL	day_sister | night_sister | staff_nurse | non_registered
ward	VARCHAR(50)	NOT NULL	Ward this nurse is assigned to

Primary Key

PK: nurseID

Foreign Keys

- ward → Ward(wardName)

Constraints

- PK_Nurse — PRIMARY KEY (nurseID)
- FK_Nurse_Ward — FOREIGN KEY (ward) REFERENCES Ward(wardName) ON UPDATE CASCADE
- CK_Nurse_Role — CHECK (role IN ('day_sister','night_sister','staff_nurse','non_registered'))

Sister

Subtype of Nurse. Represents a day or night sister assigned to a ward. Sisters supervise at the ward level and are not assigned to a specific care unit.

Columns

Column Name	Data Type	Nullable	Description
nurseID	VARCHAR(10)	NOT NULL	PK and FK to Nurse — shared identity
time	VARCHAR(5)	NOT NULL	Shift type: day or night
ward	VARCHAR(50)	NOT NULL	Ward the sister oversees

Primary Key

PK: nurseID

Foreign Keys

- nurseID → Nurse(nurseID)
- ward → Ward(wardName)

Constraints

- PK_Sister — PRIMARY KEY (nurseID)
- FK_Sister_Nurse — FOREIGN KEY (nurseID) REFERENCES Nurse(nurseID) ON DELETE CASCADE

- FK_Sister_Ward — FOREIGN KEY (ward) REFERENCES Ward(wardName) ON UPDATE CASCADE
- CK_Sister_Time — CHECK (time IN ('day','night'))

StaffNurse

Subtype of Nurse. Represents a registered staff nurse who is in charge of exactly one care unit. Implements the 1:1 manages relationship between a staff nurse and a care unit.

Columns

Column Name	Data Type	Nullable	Description
nurseID	VARCHAR(10)	NOT NULL	PK and FK to Nurse — shared identity
unitManaging	INT	NOT NULL	Care unit this nurse is in charge of

Primary Key

PK: nurseID

Foreign Keys

- nurseID → Nurse(nurseID)
- unitManaging → CareUnit(unitNo)

Constraints

- PK_StaffNurse — PRIMARY KEY (nurseID)
- FK_StaffNurse_Nurse — FOREIGN KEY (nurseID) REFERENCES Nurse(nurseID) ON DELETE CASCADE
- FK_StaffNurse_CareUnit — FOREIGN KEY (unitManaging) REFERENCES CareUnit(unitNo)
- UQ_StaffNurse_Unit — UNIQUE (unitManaging) — enforces 1:1, one nurse per unit

NonRegisteredNurse

Subtype of Nurse. Represents a non-registered nurse assigned to care for patients within a specific care unit. Multiple non-registered nurses can share the same care unit.

Columns

Column Name	Data Type	Nullable	Description
nurseID	VARCHAR(10)	NOT NULL	PK and FK to Nurse — shared identity
unitAssigned	INT	NOT NULL	Care unit this nurse is assigned to

Primary Key

PK: nurseID

Foreign Keys

- nurseID → Nurse(nurseID)
- unitAssigned → CareUnit(unitNo)

Constraints

- PK_NonRegNurse — PRIMARY KEY (nurseID)
- FK_NonRegNurse_Nurse — FOREIGN KEY (nurseID) REFERENCES Nurse(nurseID) ON DELETE CASCADE
- FK_NonRegNurse_CareUnit — FOREIGN KEY (unitAssigned) REFERENCES CareUnit(unitNo)

Patient

Represents a hospital patient assigned to one ward, one care unit, one bed, and one primary doctor. Additional treatments are tracked in PatientTreatmentPlan.

Columns

Column Name	Data Type	Nullable	Description
patientNO	INT	NOT NULL	Hospital-wide unique patient number
name	VARCHAR(100)	NOT NULL	Full name of the patient
dob	DATE	NOT NULL	Date of birth
doA	DATE	NOT NULL	Date of admission
careUnit	INT	NOT NULL	Care unit the patient belongs to
bed	INT	NOT NULL	Bed number occupied by the patient
primaryDoctor	VARCHAR(10)	NOT NULL	Doctor primarily responsible for this patient

Primary Key

PK: patientNO

Foreign Keys

- careUnit → CareUnit(unitNo)
- bed → Bed(bedNo)
- primaryDoctor → Doctor(staffID)

Constraints

- PK_Patient — PRIMARY KEY (patientNO)
- FK_Patient_CareUnit — FOREIGN KEY (careUnit) REFERENCES CareUnit(unitNo)
- FK_Patient_Bed — FOREIGN KEY (bed) REFERENCES Bed(bedNo)
- FK_Patient_Doctor — FOREIGN KEY (primaryDoctor) REFERENCES Doctor(staffID)
- UQ_Patient_Bed — UNIQUE (bed) — one patient per bed at a time

Complaint

A catalogue of medical complaints recognised by the hospital. Exists independently of any patient and is referenced by PatientTreatmentPlan.

Columns

Column Name	Data Type	Nullable	Description
complainID	VARCHAR(10)	NOT NULL	Unique complaint code (e.g. CMP001)
descr	VARCHAR(255)	NOT NULL	Human-readable description of the complaint

Primary Key

PK: complainID

Foreign Keys

None

Constraints

- PK_Complaint — PRIMARY KEY (complainID)

Treatment

A catalogue of treatments available at the hospital. Exists independently of any patient and is referenced by PatientTreatmentPlan.

Columns

Column Name	Data Type	Nullable	Description
treatID	VARCHAR(10)	NOT NULL	Unique treatment code (e.g. TRT001)
descr	VARCHAR(255)	NOT NULL	Human-readable description of the treatment

Primary Key

PK: treatID

Foreign Keys

None

Constraints

- PK_Treatment — PRIMARY KEY (treatID)

PatientTreatmentPlan

Ternary associative entity capturing the relationship between a Patient, a Complaint, and a Treatment over time. A patient may receive multiple treatments for the same complaint across different episodes.

Columns

Column Name	Data Type	Nullable	Description
patientNO	INT	NOT NULL	Partial PK + FK — identifies the patient
complainID	VARCHAR(10)	NOT NULL	Partial PK + FK — the complaint being treated
treatID	VARCHAR(10)	NOT NULL	Partial PK + FK — the treatment being applied
startDate	DATE	NOT NULL	Date the treatment episode started
endDate	DATE	NULL	Date treatment ended — NULL means still ongoing

Primary Key

PK: (patientNO, complainID, treatID)

Foreign Keys

- patientNO → Patient(patientNO)
- complainID → Complaint(complainID)
- treatID → Treatment(treatID)

Constraints

- PK_PTP — PRIMARY KEY (patientNO, complainID, treatID)
- FK_PTP_Patient — FOREIGN KEY (patientNO) REFERENCES Patient(patientNO)
- FK_PTP_Complaint — FOREIGN KEY (complainID) REFERENCES Complaint(complainID)
- FK_PTP_Treatment — FOREIGN KEY (treatID) REFERENCES Treatment(treatID)
- CK_PTP_Dates — CHECK (endDate IS NULL OR endDate >= startDate)

Normalisation

All 17 tables in this database are in Third Normal Form (3NF).

1NF: Every table has atomic columns, no repeating groups, and a defined primary key. Multivalued repeating data such as experience history and performance grades have been extracted into their own separate tables.

2NF: All non-key attributes depend on the entire primary key. In tables with composite primary keys (DoctorTeam, PastExperience, PerformanceGrade, PatientTreatmentPlan), every non-key column depends on the full composite key and not just a subset of it.

3NF: There are no transitive dependencies. For example, a consultant's specialty is stored in the Consultant table and not repeated in Doctor. Ward specialty is not stored redundantly. Every non-key attribute depends directly on the primary key and nothing else.